PERTH AI USER GROUP · JULY 2026

Architecting with AI

ArcKit — the enterprise architecture governance harness. The AI drafts. The human judges.

Chris McKelt
Enterprise Solution Architect · Smart Tech Ventures

arckit.org

ABOUT ME

A practitioner's view, not a pitch

Chris McKelt — 25+ years across AI, cloud, digital and data. Multi-cloud certified, independent, based in Western Australia.

About Me

mckelt.com

My background, CV and past project

Smart Tech Ventures

smarttechventures.au

AI architecture and advisory consulting business I own.

Cohezion

cohezion.com.au

Delivery partner engaged on a case study outlined in the talk

linkedin.com/in/chrismckelt   github.com/chrismckelt   chris@mckelt.com

TONIGHT

Overview

WHAT

What is ArcKit

- Open-source, AI-assisted architecture toolkit
- Governed, traceable artefacts
- Strategy, discovery, requirements, risk & solution design

WHY

Why it exists

- Architecture work is slow, fragmented, hard to govern
- Accelerates delivery
- Keeps rigour, traceability & human oversight

WHO

Who is it for

- 18 role guides across the organisation
- Australian sector overlay

HOW

How to use it

- Real consulting engagements
- End-to-end demo: a fictional university transformation

WHAT · ARCKIT

A governance harness for AI assistants

ArcKit turns an AI assistant into a governed document engine. Deliverables are generated by slash commands, version-controlled, stamped with provenance and reviewed by humans before reliance.

Open source

MIT-licensed toolkit by Mark Craddock (tractorjuice). Docs, guides and 40+ public example projects at arckit.org.

Runs inside your AI assistant

Plugs into Claude Code, Codex, Gemini, Copilot or OpenCode — you bring the AI you already pay for; ArcKit is workflow, not inference resale.

Markdown + Git, not documents

Artefacts live in the repo: versioned, diffable, reviewable via pull request. Governance as code.

Governed, not just generated

A harness intercepts every write: gates block unsafe changes, observers stamp provenance, traceability links end-to-end.

Inspired by GitHub's Spec Kit — adapted from software specs to enterprise architecture governance.

WHAT· THE HARNESS

The harness: AI drafts; humans review & decide

1
AI assistant
Claude Code · Codex · Gemini · Copilot

→

2
Gates
Policy checks before write

→

3
Commands + overlays
Templates, AU frameworks, sector packs

→

4
Observers
Provenance + traceability stamps

→

5
Git artefact
Versioned, diffable, reviewable

71	16	128	12
COMMANDS	AGENTS	DOC TYPES	JURISDICTION OVERLAYS

Pre/Post hooks ensure traceability end to end

WHAT · REFERENCE ARCHITECTURE

The whole system on one page

GOVERNED ARTEFACTS

every document is a linked node in the traceability graph

Stakeholders · Goals · Requirements · Decisions · Risk register · Data model · Diagrams · Wardley maps · Business case · Roadmap · Privacy · +60 more

↑ generates — provenance-stamped + traceable

THE ENTERPRISE GOVERNANCE HARNESS

OVERLAYS

9 sector / jurisdiction packs — Australia, UK NHS, UK Finance, EU, USA, Canada & more

ARCKIT CORE ENGINE

71	16	5	128
commands	agents	skills	doc-types

MCP KNOWLEDGE

Live external grounding — AWS, Microsoft Learn, Google dev, Data Commons

hooks fire at every lifecycle stage: PreToolUse gates · PostToolUse observers · UserPromptSubmit · SessionStart · provenance-stamp · secret-file-scanner

↑ tool calls — Write / Edit / Bash …

AI CODING ASSISTANTS

Claude Code · Codex · Gemini · OpenCode · Copilot · CLI

WHAT · TRACEABILITY

/arckit:pages

Dependencies as a data structure

Category-layered layout

Documents grouped by governance domain, edges drawn from the traceability graph.

Orphan detection

Dashed borders reveal documents nothing links to — governance debt made visible.

Freshness indicators

Green under 7 days, amber 7–30, grey stale — staleness tracked per artefact.

Interactive exploration

Hover highlights the dependency chain; click navigates to the artefact.

14 nodes **11** edges **5** categories **2** orphans built into ArcKit pages · zero dependencies

WHAT · COVERAGE

One toolkit, many methods

Wardley Mapping	Heaviest by far — 5 dedicated commands: value chain, map, climate, doctrine, gameplay
HM Treasury Green Book	`sobc` — five-case business model
Platform Design Toolkit	`platform-design` — 8 canvases
C4 model	`diagram`
TOGAF ADM	Separate plugin — 9 commands, Preliminary through Phase H

GDS Service Standard	`service-assessment` · `tcop`
HM Treasury Orange Book	`risk` — scored register with owners and mitigations
ITIL	`servicenow` · `operationalize`
Yourdon-DeMarco	`dfd`
Spec Kit	Handoff documented in the enterprise-scale guide

OVERLAYS 12 bundled sub-plugins: uk · us · eu · fr · ca · au · uae · at, plus agent, fde, repo, togaf. **You have au installed.**

WHAT· SCALE WITH AGENTS

`/arckit:build` — from sequential grind to parallel waves

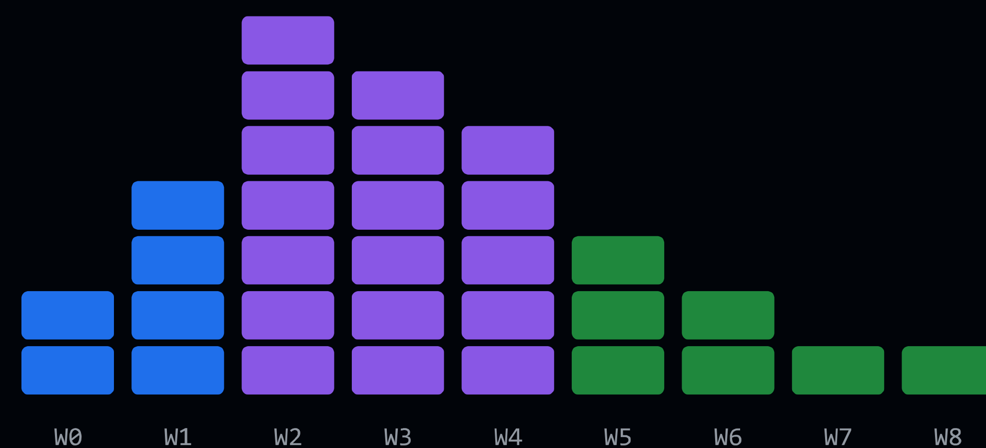
31 artefacts in one run

9 dependency-ordered waves

1 git commit per wave

Recipe-driven YAML. One subagent per artefact per wave, each in fresh context. Resumable via `state.json`.

`Run /arckit:build 001 --plan`



Artefacts per wave — uk-saas recipe wave plan

WHY · THE ECONOMIC CASE

Australia overspends on the Big 4 — and is pulling back

Australian government consulting spend, 2024–26. A billion-dollar drafting layer is being repriced in real time.

A$1bn+

Federal government's annual consulting bill to its biggest firms

The Australia Institute, via Consultancy.com.au

A$890M

Federal cut to Big Four consulting spend over two years, from late 2024

Mordor Intelligence

↓ 36%

Big Four federal contracts FY26 YTD (A$415M) against the prior year

AusTender, via Psithur

In-house

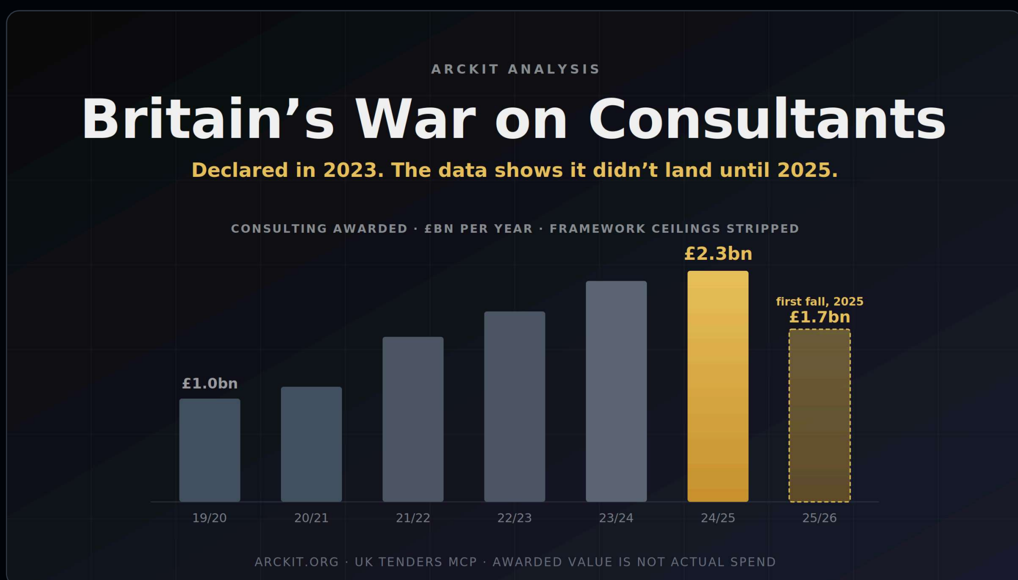
Australian Government Consulting (AGC) stood up to rebuild advisory capability inside the APS

Consultancy.com.au

Generic advisory drafting is going in-house. **And Australia is not the first — the UK hit this wall two years earlier.**

WHY · THE SAME PATTERN ABROAD

The UK hit the same wall — and reached for ArcKit



Same overspend, bigger numbers

The “war on consultants” was declared in 2023, yet awarded value kept climbing to £2.3bn — the first real fall came only in 2025.

ArcKit is the UK’s answer

Born in UK public-sector delivery and adopted within several sectors — central government, NHS and finance overlays ship in the toolkit.

Architecture at the speed of the decision

AI coding assistants plus the ArcKit harness turn the repeatable £250k–£2M phase-zero pack into governed artefacts delivered in days, not months.

WHY · THE PROBLEM

In the past...

Scattered documents, no single source of truth

Requirements, risks, decisions and designs live across SharePoint, Confluence, email and slide decks — and drift apart.

Weeks of expert effort per document

Business cases, risk registers and reviews queue behind delivery — the expert-drafting layer is what everyone waits for.

No answer when the auditor asks why

Nothing links a decision back to a requirement, a risk, a business driver. The audit trail is reconstructed from memory.

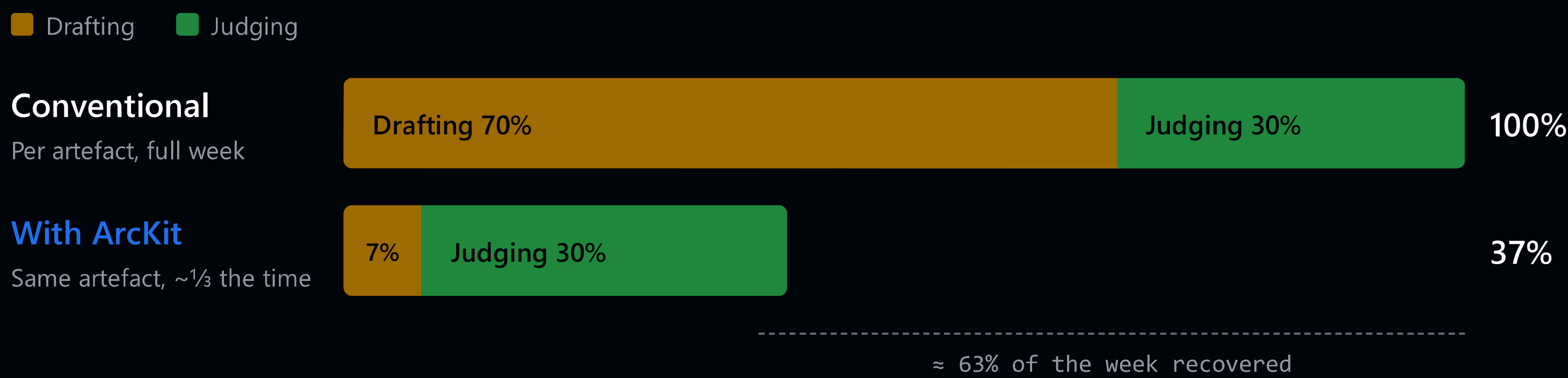
Raw GenAI makes it worse

Fast, plausible drafts — no provenance, no gates, no audit trail. Speed without control is a new liability, not a productivity gain.

The question is not "can AI draft it?" **— it is "can I defend it?"**

WHY · THE WORKING DAY, REWRITTEN

Drafting collapses; judging stays constant



70→7%
DRAFTING SHARE

2.5×
DELIVERY THROUGHPUT

0
CHANGE IN ACCOUNTABILITY

Drafts ratified, not retyped
The architect's signature stays

WHO · ROLES

arckit.org/roles

18 role guides — start from your job title

ARCHITECTURE	Enterprise Architect 12 · Solution Architect 10 · Technical Architect 5 · Business Architect 5 · Security Architect 5 · Data Architect 4 · Network Architect 3
CHIEF DIGITAL & DATA	CTO / CDIO 5 · CISO 5 · CDO 4
PRODUCT & DELIVERY	Delivery Manager 6 · Product Manager 5 · Business Analyst 4 · Service Owner 3
DATA	Data Governance Manager 4 · Performance Analyst 4
OPS & DEVELOPMENT	IT Service Manager 3 · DevOps Engineer 3

DDAT-MAPPED Each guide maps primary commands onto the UK Government DDaT Capability Framework — the number is commands per role; depth tracks breadth of remit, 12 for Enterprise Architect down to 3 for the most focused roles.

WHO · AUSTRALIAN MARKET

Compliance as commands — the Australian overlay

Ten community-contributed commands anchored on Australia's federal frameworks. Each produces a governed, gated, traceable artefact.

`/arckit:au-e8-posture`
ASD Essential Eight maturity

`/arckit:au-ism-controls`
ISM Statement of Applicability

`/arckit:au-pia`
Privacy Act 1988 PIA, APP-by-APP

`/arckit:au-ndb-playbook`
OAIC Notifiable Data Breaches

`/arckit:au-dss`
DTA Digital Service Standard

`/arckit:au-pspf`
PSPF protective security

`/arckit:au-ai-assurance`
DTA AI Assurance Framework

`/arckit:au-disp-attestation`
Defence Industry Security Program

SECTOR PACKS OT / ICS / SCADA security · SOCI Act / CIRMP evidence · AESCSF energy-sector cyber maturity — composed on the AU federal baseline. Further community overlays: EU · UK · Canada · UAE · USA.

CASE STUDY 1

WA community services — enterprise transformation

Architecture and governance to design the future operating state across five projects — IDAM, security, payroll, customer management, HR, learning and finance.

109
governed architecture artefacts across five projects

342
traced business, functional, NFR and integration requirements

96%
project traceability coverage

70
risks, every one with an owner and mitigation before contracting

Vendor profiles, weighted evaluation frameworks, pre-RFP scorecards, a Statement of Work, regulatory deep-dives — and all 19 architecture principles conformant at design stage.

`PLAN → STKE → REQ → RISK → DATA → SOBC → HLD → HLDR → ADRs → AUPIA → TRAC → CONF · 30+ governed artefacts, versioned and traceable`

CASE STUDY 2

ASX-listed mining technology — data mesh platform

Architecture and governance for a global cloud data mesh — Snowflake, Azure/AKS and MLOps services across nine product domains and a central BI function.

87/100

governance health score

100%

design coverage for all 95 traced requirements

13

Architecture Decision Records, plus HLD and independent review

~25

governed artefacts in one version-controlled corpus

C4 diagrams, a 27-entity data model, reusable data-contract standards, MLOps guidance — plus an Orange Book-style risk register, STRIDE threat model, Essential Eight baseline and privacy assessment.

ArcKit delivered: a baseline auditable by construction — and explicit about the work required before build.

Two engagements, one governed process

Different assistant, different security posture — same ArcKit workflow.

PSPF-style classification

- Every artefact carries a PSPF marking (UNOFFICIAL → PROTECTED)
- Handling caveats as document metadata
- Enforced by configuration

Assistant-agnostic in practice

- One engagement on OpenAI Codex CLI, one on Claude Code
- Same artefacts, same governance
- No lock-in to one model

AU assurance by default

- Essential Eight posture & ISM controls
- Privacy Act PIA + NDB playbook
- Applied wherever personal information or ASD baselines apply

Meets stakeholders where they work

- Client-local skills extended ArcKit
- One-command export to Azure DevOps Wiki & Confluence
- Governance went to the reader

The AI drafts; the human judges

Drafting collapses; judgement doesn't

- Drafting time falls to near zero
- Stakeholder context & trade-off thinking stay constant
- The expert's day shifts from writing to reviewing

Traceability changes conversations

- Every requirement links to a driver; every decision to a requirement
- Procurement disputes, audits & scope debates resolved from the artefacts — not memory

Overlays make compliance a starting point

- Privacy, cyber & AI-assurance commands run early
- Obligations surface while the architecture can absorb them cheaply — not as findings at the end

AI output still needs qualified owners

- Overlays are accelerators, not opinions of record
- Security, privacy & legal owners review before reliance
- The harness makes review easy, not optional

DEMO · THE SCENARIO

The University of Funk — a fictional Australian university

Mid-sized & specialised

A fictional mid-sized Australian university with strong Music, Performing Arts and Health Sciences faculties.

Ten years of organic growth

The Learning & Teaching technology ecosystem grew tool by tool — no central plan, each faculty solving its own problem.

The result

Overlapping tools, fragile integrations and inconsistent governance across the whole L&T estate.

https://github.com/SmartTechVentures-au/arckit-demo

DEMO · THE PROBLEM

20+ platforms, no single owner

LMS · Lecture capture · Assessment · Placement · Collaboration · Analytics · + 14 more

Duplicated capability

Multiple tools doing the same job — and licence value paid for but never used.

Fragile integrations

Point-to-point connections and manual workarounds holding critical processes together.

Privacy & ownership

Student data spread across platforms with privacy exposure and unclear platform ownership.

DEMO · THE COMMISSION

A defensible roadmap and a business case to optimise

UoF has commissioned a **Learning & Teaching Baseline Strategy** to rationalise the ecosystem.

1. Map current systems to capability
2. Apply academic requirements
3. Define architecture principles
4. Assess risk
5. Produce a defensible roadmap and a business case to optimise the ecosystem

Live demo from here — fully live, inputs are the only pre-work. Recovery: `/arckit:health`, or reset to a checkpointed `demo/stage*` branch.

Governance at the speed of AI.
Audit-ready by construction.

1 · EXPLORE

`arckit.org`

Guides, playbooks and 40+ public example projects across health, AI, cloud and data.

2 · INSTALL

```
claude plugin install arckit arckit-au
```

The core harness plus the Australian overlay. Free, open source.

3 · PILOT

`/arckit:init`

Pick one live initiative, follow the navigator, and put the eight essential artefacts in Git within a fortnight.

ArcKit is MIT-licensed open source by Mark Craddock (tractorjuice) · github.com/tractorjuice/arc-kit. Community AU overlays are accelerators — review with qualified security, privacy and legal owners before reliance.

Thanks

MC

Mark Craddock

Lead ArcKit Architect & FDE Specialist

ArcKit · United Kingdom

- Architecture governance
- AI-assisted delivery
- Team training
- Bootstrap engagements

Creator of ArcKit; leads the Forward Deploy Engineering practice. Embeds senior architecture capability directly into delivery teams to produce decision-grade evidence at pace.

CM

Chris McKelt

Enterprise Architect & Forward Deployed Engineer

Smart Tech Ventures · Perth, WA

- Enterprise architecture
- Cloud & data
- Governed enterprise AI
- ArcKit training

25+ years across enterprise architecture, delivery and technology leadership. Applied ArcKit across two client engagements with a third starting at a university; contributes Confluence, Azure DevOps Wiki and Jira exporters plus the dev-container template back to the project.

smartechventures.au/arckit linkedin.com/in/chrismckelt github.com/chrismckelt chris@mckelt.com